Code to Connect | S&P Global & Crisil | Hackathon 2026

- **Problem Statement Title- RiskPulse: real-time financial risk intelligence from unstructured text**
- **Name: Raunak Kumar Jha**
- **College: Indian Institute of Information Technology (IIIT) Kottayam**
- **Email: raunakjha24bcy27@iiitkottayam.ac.in**
- **News and social posts in, evidence-backed risk signals and portfolio stress tests out.**



RiskPulse: Real-Time Financial Risk Intelligence

From unstructured text to evidence-backed risk signals and portfolio stress tests

S&P Global Crisil
a company of S&P Global

IDEA/SOLUTION :

How we turn a flood of headlines into a risk decision?



Understand & Connect

Collects financial news (RSS) and Reddit posts, and links each text to S&P 100 companies by name, ticker or cashtag.

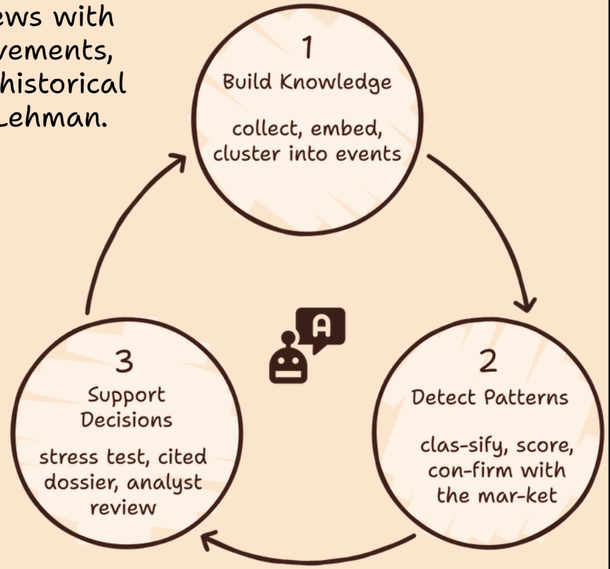
Drops duplicates through a semantic cache, and every output follows one shared JSON contract

Research & Correlate

Groups similar articles into a single event, counts independent sources, correlates news with price and volume movements, and compares it with historical events like SVB and Lehman.

Predict & Guide

Scores sentiment (-1 to 1), event type, impact (1 to 10) and confidence, and escalates only uncertain or high-impact items to an LLM
Stress-tests the portfolio for loss, VaR, expected shortfall.



- **Fast first, smart when needed:**The LLM runs only when confidence is low or impact is high, and it must back its answer with a quote that exists in the source.
- Uses **Redis semantic cache** to reuse responses for similar queries, reducing repeated retrieval and LLM calls while improving response speed and system efficiency.

UNIQUENESS:

- **Evidence-Locked AI:** The LLM's quote must appear verbatim in the source, and a second opinion must agree, or the item goes to human review instead of being guessed.
- **Corroboration Engine:** Signals are grouped into events and judged by independent source count and market confirmation, not by a single viral post.
- **Degrade, Don't Die:** If the LLM or a news source fails, the pipeline keeps emitting signals marked degraded. You can inject the failure live from the dashboard.

Parallel AI Research

AI agents can research multiple documents simultaneously, saving time.



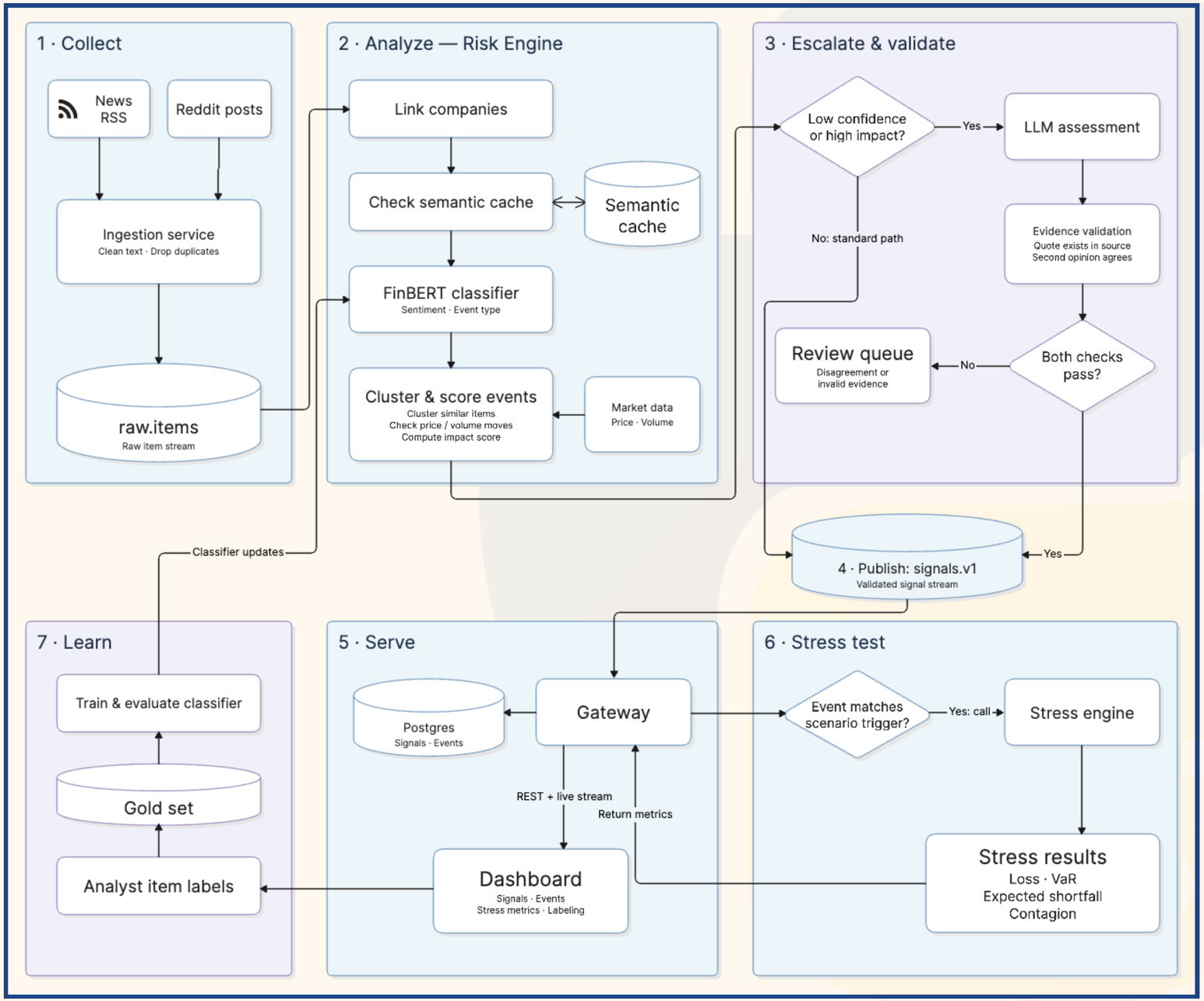
Deep Research

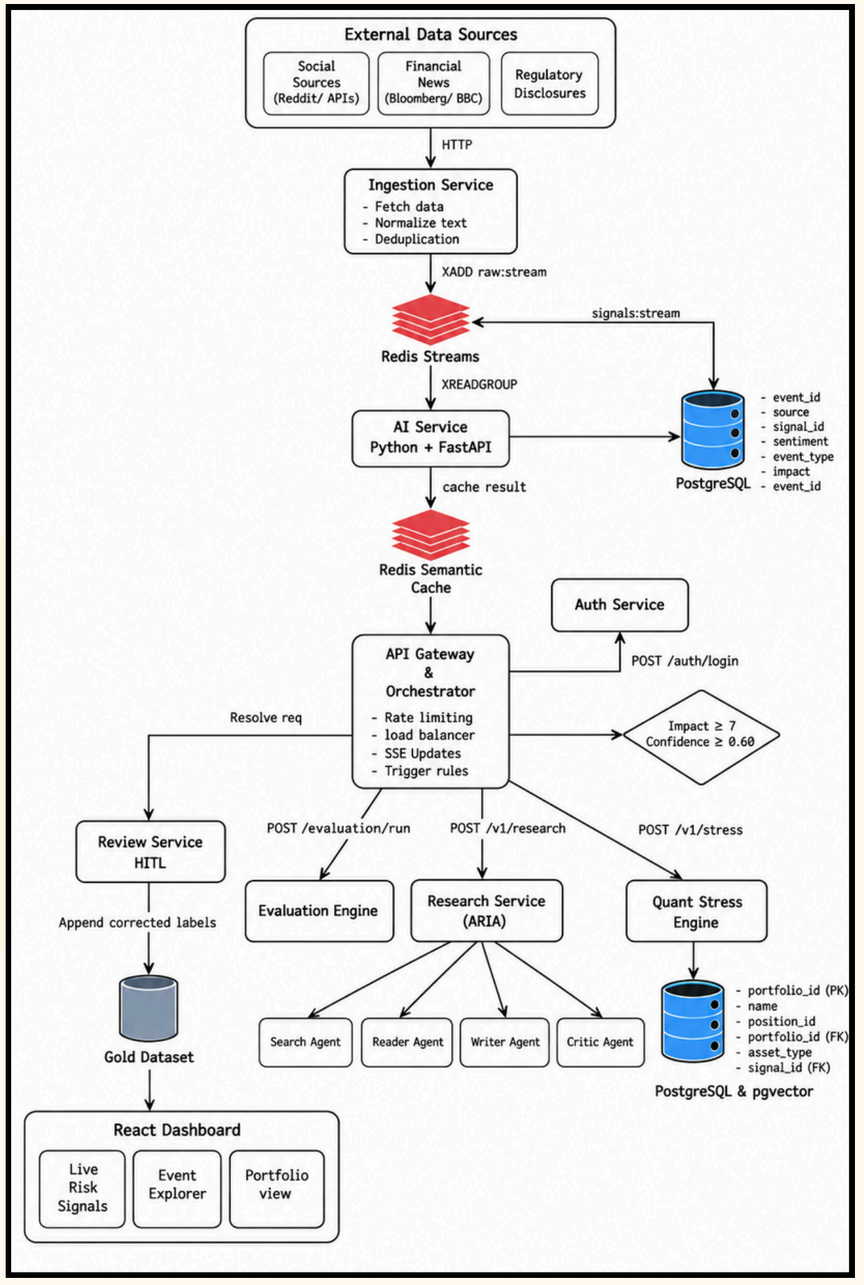
Users can perform in-depth research on specific terms or clauses.

- **History-Anchored Impact:** Hybrid retrieval (BM25 plus embeddings) finds similar past events like SVB and Lehman, and their real price reactions anchor the impact score.
- **Self-Learning Loop:** Items the system won't score go to a review queue, and analyst labels flow into the gold set that trains the classifier.

System Design

S&P Global
Crisil
a company of S&P Global





Implementation Highlights

S&P Global Crisil
a company of S&P Global

Data Processing and Retrieval

- **Event Clustering:** Uses **BGE embeddings** and **pgvector** to group similar news articles into one event and count independent sources.
- **Historical Retrieval:** Combines **BM25, BGE embeddings, and rank fusion** to etrieve relevant exampes from 64 historical market events.

Model Optimization and Efficiency

- **Semantic Cache:** Uses company-scoped caching and a meaning-flip guard to **avoid repeated model calls** without reusing incorrect answers.
- **Model Cascade:** Uses **LangGraph** to run fast models first and invoke LLMs for uncertain cases, **reducing cost and latency.**

Verification and Security

- **Verified LLM Output:** Checks whether generated quotations exist in the original sources and uses a second opinion to reduce fabricated evidence.
- **Security Guardrails:** Implements **prompt-injection filtering**, schema validation, and a 58-input adversarial test suite.

Research and Analysis

- **ARIA Research Agents:** Uses a four-agent pipeline to generate cited event dossiers covering claims, counter-evidence, historical analogies, and portfolio exposure.
- **Stress Engine:** Performs portfolio repricing, con-tagion analysis, **VaR, Expected Shortfall (ES)**, what-if analysis, and reverse stress testing.

System Improvement and Reliability

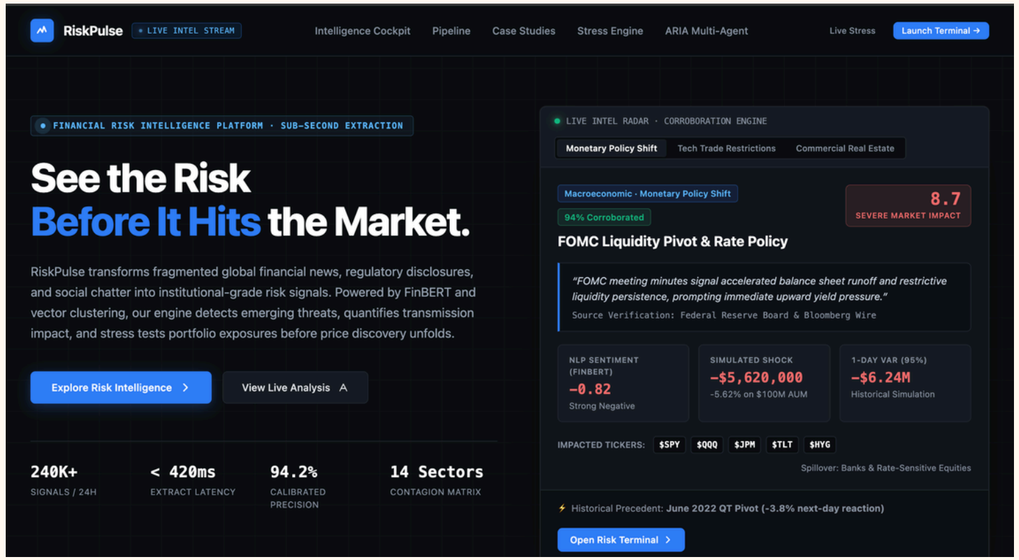
- **Self-Learning Loop:** Uses analyst **feedback, labelled examples, a gold-standard dataset,** and classifier retrain-ing to **improve predictions.**
- **Reliability Engineering:** Implements **retries, circuit breakers,** dead-letter queues, **replay mechanisms,** and fault-injection testing.

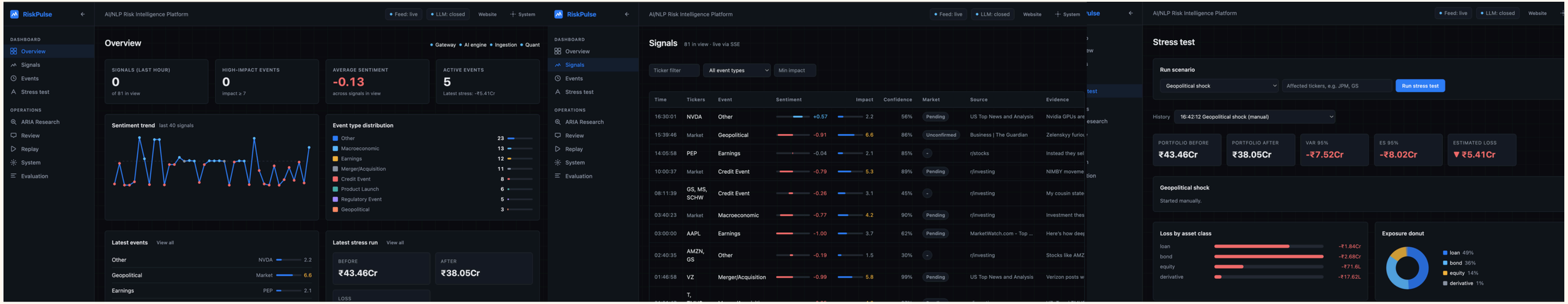
Tech Stack:

Key results: measured, not claimed

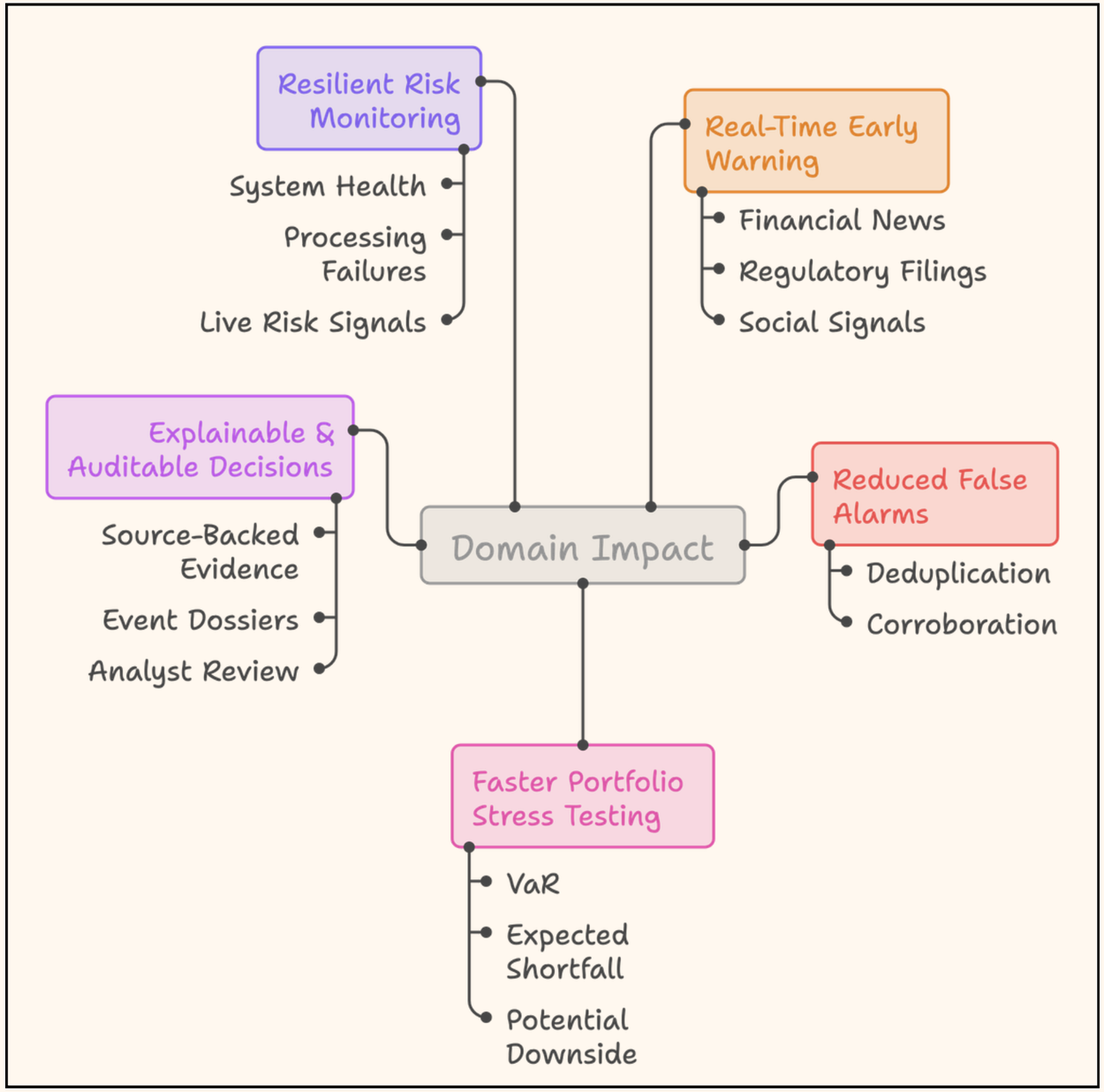
S&P Global
Crisil
a company of S&P Global

Result	Value	Footnote
Sentiment accuracy (FinBERT)	97.4%, macro-F1 0.964 (95% CI 0.952 to 0.974)	Measured on Financial PhraseBank. FinBERT was fine-tuned on it.
Confidence calibration	Calibration error 0.058 to 0.007 after temperature scaling, a 88% reduction	Temperature 0.60, fitted on a held-out 30% and measured on the other 70%
Embedding choice	BGE separates paraphrases from meaning flips with AUC 0.98. MiniLM scores 0.36	14 hand-built tricky pairs, so it is a small sample
Hostile input defenses	48 of 48 component checks passed	58-input adversarial suite





Domain impact



Real-Time Early Warning: Monitor financial news, regulatory filings, and social signals to detect emerging risks early.

Reduced False Alarms: Deduplicate repeated reports and corroborate events across independent sources.

Faster Portfolio Stress Testing: Translate major events into portfolio shocks and assess VaR, Expected Shortfall, and potential downside.

Explainable & Auditable Decisions: Provide source-backed evidence, event dossiers, and analyst review with feedback capture.

Resilient Risk Monitoring: Track system health, handle processing failures, and maintain visibility into live risk signals.

Multi-Agent Research Intelligence: Coordinate specialized AI agents to search financial sources, extract evidence, cross-verify claims, and generate citation-backed research reports.

Limitations and next steps

S&P Global
Crisil
a company of S&P Global

ASSUMPTIONS

- **Instantaneous Shocks:** Rates, credit spreads and equity factors shift immediately during stress scenarios.
- **Source Corroboration:** Social signals remain unconfirmed until supported by multiple independent sources.
- **Mapped Universe:** Entity extraction focuses on the S&P 100; other entities map to sector benchmarks.
- **Static Portfolio:** Positions remain unchanged during the initial stress window.

CURRENT GAPS

- **English-Only Ingestion:** Limited multilingual financial news and regulatory coverage.
- **SLimited Gold Data:** Rare corporate restructuring and private-debt events need more labelled examples.
- **No Contagion Cascades:** Interbank counterparty defaults and liquidity spillovers are not fully modelled.
- **No Order-Book Data:** Tick-level depth and options implied volatility are not integrated.

FUTURE WORK

- **Graph Neural Networks (GNNs):** Represent interbank lending and counterparty relationships as a graph to identify interconnected institutions and predict cascading defaults and credit contagion.
- **Cross-Lingual NLP:** Integrate multilingual language models to analyze financial news, central bank announcements, and regulatory disclosures across languages, expanding coverage of global markets.
- **Active Learning:** Use analyst feedback to improve model quality through controlled retraining.
- **Automated Hedging:** Integrate with broker systems through the FIX protocol to generate risk-based hedging orders, subject to position limits, pre-trade risk checks, and human approval.